

JeePS 2.0: Enhancement of a Real-time Transportation Tracking System for PUVs

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ABSTRACT

JeePS, a real-time transportation tracking system, is a web and mobile application designed to improve passenger satisfaction with Public Utility Vehicles (PUVs). The features within the application were designed to address key factors like safety, driver's behavior, service adequacy, and information materials, while also solving user transportation problems. However, there remains room for improvement, as certain limitations and issues have been identified in the system. These include security vulnerabilities, a limited feedback and reporting system, GPS service constraints, an inefficient passenger counter, and insufficient user testing.

This paper introduces a design of the JeePS version 2.0, which enhances security, safety, and overall system efficiency. A domain expert validated its design and confirmed that it covers the necessary features to address the limitations of its original version, including enhancements to the feedback and report system, passenger demand summary, and integration of an infrared (IR)-based automatic passenger counting (APC) system. Preliminary hardware testing shows that the APC performs well in detecting single and multiple passengers but needs improvement to handle complex scenarios like slow-moving, hesitant, and simultaneous passengers entering or exiting the jeepney. To ensure that the application meets its purpose to improve passenger satisfaction and service quality of jeepneys, plans for functionality and field testing are presented.

CCS CONCEPTS

• **Computer systems organization** → **Real-time system specification; Sensors and actuators;**

KEYWORDS

service quality, passenger satisfaction, automated passenger counter (APC), public utility vehicle (PUV), jeepney

1 INTRODUCTION

The 2023 Urban Mobility Readiness Index ranked Manila, the capital of the Philippines, among the worst cities for public transit systems across the world, which indicates high public transit density, low efficiency, and poor utilization of mobility services [1]. To address the inadequate state of public transportation, it is crucial to closely examine PUVs as they make up a significant portion of daily commuting options. Jeepneys, a primary example of PUV, dominate informal public transportation due to their flexible operation. However, little is known about informal transport operations causing difficulties in improving the transportation system, which negatively influences the service quality provided by jeepneys [7].

In a study by Ong et al. [16], service quality dimensions relevant to jeepneys were used to identify the factors that mostly affect passenger satisfaction. Findings from this study state that the government and jeepney drivers can improve service quality by focusing on these specific dimensions. The main factors include safety, value for money, driver's behavior, information materials, and service adequacy. These also align with other factors like customer care, information provision, and reliability identified in related studies on service quality perception on public transportation, particularly in paratransit systems [19][18]. Collectively, these studies support the idea that meeting quality standards in these areas is essential for improving the passenger experience and service quality for a convenient and accessible public transportation system.

1.1 Solutions Made to Address Public Transportation

In a study by Goletz et al. [7], the use of Global Positioning System (GPS) can be used to better understand informal transportation, such as jeepneys, to enhance their service quality and improve passenger satisfaction. Several solutions utilizing GPS have been developed to address the challenges in public transportation in the Philippines.

Sakay.ph is one of the more famous transport management systems designed to assist operators in planning, creating, and editing routes while generating reports and analytics. It offers different modes of transportation and features selected real-time GPS tracking, payment, ticketing, and booking systems [5]. According to the assessment of Toleda et al. [21], Sakay.ph effectively allows users to customize their travel preferences, optimize routes, and access a commuting guide. However, it scored below average in the overall satisfaction in terms of usability and failed to accommodate users in navigating routes based on specific time schedules.

Another solution is iBiyahé which consists of a web application for administrators and a mobile application for both drivers and commuters. It provides an account system, real-time location tracking of PUVs, route selection, real-time traffic information, PUV availability updates, and notifications. The system provides a user-friendly interface that helps commuters by reducing the difficulty of finding available PUVs, displaying accurate real-time locations, and offering relevant traffic information. Drivers also find it easy to navigate and useful for increasing the number of passengers. While the administrators can monitor PUVs and drivers, iBiyahé could benefit from additional features to enhance management and monitoring capabilities [6].

Similarly, JeePS is a transportation tracking system composed of a web application for passengers and route managers and a mobile application for drivers. Its key features include real-time GPS

tracking, a fare matrix, a passenger counter, rough time estimation, a feedback system, a reporting system, a passenger demand system, and an account system [17].

1.2 JeePS

Although Sakay.ph and iBiyahe offer valuable functionality, JeePS provides features that directly address the factors affecting passenger satisfaction and user transportation problems. Unlike other systems, JeePS includes a feedback and report system that addresses safety, which is the number one factor passengers are concerned about in jeepney transportation [16]. In addition, its passenger demand system keeps users engaged by providing drivers with real-time passenger demand to ensure commuters can rely on the application for efficient travel. This feature allows passengers to mark their location on the map, allowing drivers to see waiting passengers within the map.

User testing results on JeePS validated its effectiveness, demonstrating a 33% average increase in passenger satisfaction. Additionally, the system received a score of 87.5% on the System Usability Scale (SUS) reflecting a user-friendly design crucial for both commuters and drivers navigating the app while on the road.

Building on the foundations of JeePS, this paper presents JeePS 2.0, an enhanced version that addresses the limitations of the original application. In the next section, the authors evaluate the previous version to establish what improvements are needed for the design, along with the new and improved software features and the hardware design to support the system. Additionally, the initial hardware testing results for the IR-based APC implementation are also presented in Section 4.

To assess whether the enhancement of JeePS 2.0 is significant for improving the passenger satisfaction and service quality of jeepneys, a testing plan for both application versions is discussed in Section 5.

Moreover, Section 6 presents expert validation from a transportation and technology specialist at the National Center for Transportation Studies (NCTS), including general feedback on the system's features, while Section 7 provides the overall conclusion and recommendations for the project's design.

1.3 JeePS Limitations

Despite the success of JeePS, certain limitations have been identified based on user feedback and recommendations. The following outlines these limitations and potential areas for improvement:

- (1) **Limited Feedback and Report Context** - Without image attachments, feedback and reports may have insufficient contextual information, especially when reporting accidents or lost items. Moreover, the system operates as one-way communication, preventing users from knowing if their reports are acknowledged.
- (2) **System Integrity** - There is no way of moderating the feedback and reports sent by users to prevent abuse, such as spamming false reports or submitting troll feedback. According to Gosman et al. [8], the reliability of the transportation system is undermined without controlling or filtering the data shared by the user. Data submitted by malicious participants regarding jeepneys, routes, or drivers may contradict

the observations of the majority, which can lead to inaccurate information and hinder the system's effectiveness.

- (3) **Passenger Demand View** - Passenger demand data is only visible in the route manager's web application and the driver's mobile application instead of being visible to passengers. Without this information, perceived waiting time is affected, influencing passenger satisfaction [4].
- (4) **Passenger Demand Data** - Passenger demand data is exclusively available to the route manager, and remains unutilized for potential applications, like in short-term passenger demand forecasts [10].
- (5) **Limited Fare Matrix** - Fare prices displayed only include base prices for the routes, leaving passengers without information on pricing adjustments based on varying distances and discounts.
- (6) **Unreliable Location Updates** - The jeepney's location does not update when the mobile application is running in the background or when the driver's mobile phone is locked, which may impact real-time tracking accuracy.
- (7) **User Data Exposure** - The driver's location may unintentionally be exposed when the app is left open even when not operating. According to Gosman et al. [8], this is a potential for misuse as malicious users could exploit the GPS data to deduce location patterns and geographical habits.
- (8) **Inefficient Passenger Counter** - The passenger counter was implemented through a manual counting mechanism where drivers must manually click a button through their mobile application whenever passengers enter or exit the jeepney. However, user test results have shown that this implementation was too cumbersome for drivers.
- (9) **Passenger Safety** - With limited GPS tracking capabilities, users cannot track the current location of other passengers of their concern, similar to the location-sharing feature found in ride-sharing platforms [3].
- (10) **Limited User Testing** - Given the time constraints and communication bottlenecks with the jeepney operators, the JeePS developers only had one day for the live testing, resulting in limited results.

2 OBJECTIVES

This research aims to enhance passenger satisfaction and the service quality of jeepneys. Specifically, it focuses on addressing the issues and limitations of the JeePS system by improving its software component and integrating a new hardware component for automating passenger count. Additionally, it seeks to gather more insights into JeePS by implementing a rigorous testing plan, enabling a comprehensive performance assessment and comparison with JeePS 2.0.

3 SYSTEM DESIGN

The system consists of a mobile application for drivers and a web application for route managers and passengers as illustrated in Figure 1. The passenger, route manager, and driver each have their respective accounts within the system, while a passenger who is not logged in is considered a guest user. Only the drivers interact

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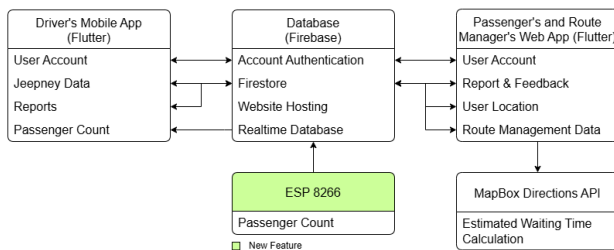


Figure 1: System Architecture

with the mobile application, while the route managers and passengers use the web application. Route managers have access to all passenger features with other functionalities that allow moderation of the information displayed within the web app.

User accounts are managed securely through Firebase's authentication system, while all other data required by JeePS is efficiently stored and managed using Firebase Firestore and Realtime Database.

The ESP8266 microprocessor plays a key role in updating the passenger count in Realtime Database. Additionally, the MapBox Directions API powers the system's estimated waiting time calculation feature.

3.1 Software Implementation

JeePS 2.0 retains all the key features from its previous versions, ensuring continuity in functionality. The following are the features that have been carried out from the earlier versions.

- (1) **Account System** - Ensures that access to specific features by logged-in users is regulated. While users who are not logged-in can still access the web application, access to the Passenger Demand System, Feedback System, and Report System is restricted.
- (2) **Basic Information** - Shows passengers the essential details, such as the driver's name and jeepney plate number in the web application.
- (3) **Rough Time Estimation** - Estimates the time each jeepney will take to arrive at the passenger's location which serves as a rough estimate of the waiting period visible through the web application.

From the existing features, several improvements have been made to the following features to further enhance user experience and data privacy.

- (4) **Real-Time GPS Location** - Displays the live location of jeepneys on the map, even if the driver's phone is locked or in the background, addressing unreliable location updates of PUVs. This also allows drivers to share their location without having their phone open for the whole trip.
- (5) **Passenger Counter** - Integrates a hardware-based APC, explained further in the next section, which shows real-time occupancy in the web and mobile applications that addresses manual counting issues.
- (6) **Feedback System** - Allows logged-in passengers in the web application to view and provide feedback on the driver

and their respective jeepney through a 5-star rating system and comments with the option to attach images for additional context. Route managers can also moderate feedback as shown in Figure 5 to ensure system integrity. They now have the option to remove troll or inappropriate feedback, ensuring that only relevant and constructive feedback remains visible. This helps create a more reliable feedback system and fosters a better community environment for users.

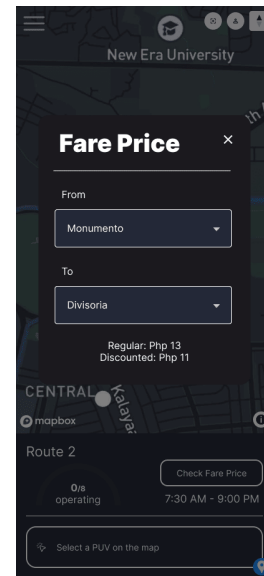


Figure 2: Fare Matrix UI

- (7) **Fare Matrix** - Allows passengers to view fare prices based on the usual stops within the route, extending the fare matrix in the web app shown in Figure 2.

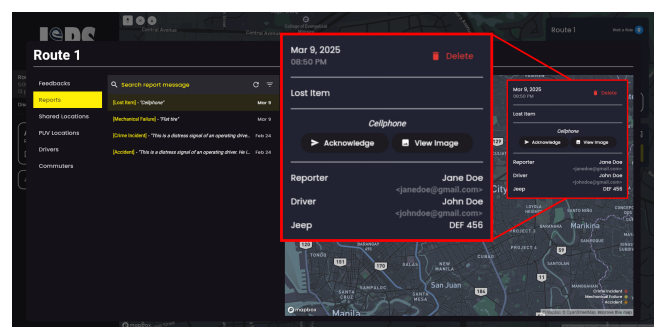


Figure 3: Report System UI

- (8) **Report System** - Allows drivers to report crimes, accidents, or mechanical failures from the mobile application directly to the users in the web application. In addition to the driver options, logged-in passengers can also report lost items and miscellaneous concerns along with brief descriptions and optional image attachments in the web application. This feature

alerts both the drivers and route managers on their maps. Moreover, route managers can acknowledge and moderate reports as shown in Figure 3 to improve response efficiency and maintain system integrity.

- (9) **Passenger Demand System** - Enables logged-in passengers to mark their location on the map, making their location visible to the drivers in the mobile application. In addition, logged-in passengers can also view other waiting passengers on the map through the web application, improving perceived waiting times.

Building from the success of its earlier versions, JeePS 2.0 introduces the following new features designed to further improve passenger satisfaction and enhance operational efficiency of jeepney drivers.

- (10) **Live Location Sharing** - Passengers with an account in the system have the option to share their live location and jeepney details to family members or friends while they are commuting. This effort, which has been successfully implemented by the Grab application for commuting on private vehicles, is adopted in JeePS 2.0 to offer additional measures for enhancing passenger safety [9].

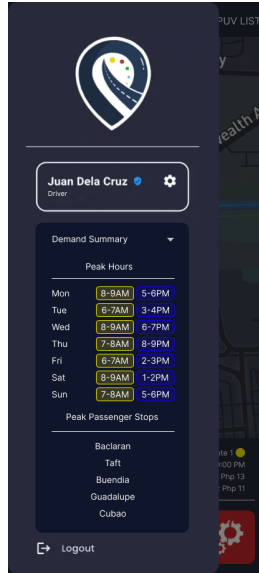


Figure 4: Passenger Demand Summary UI

- (11) **Passenger Demand Summary** - Figure 4 shows a summarized view of times and locations along the jeepney route with high passenger demand based on historical data from pinged passenger locations. This helps drivers using the mobile application make informed decisions on scheduling their trips and avoid idle times that may reduce fuel consumption and maximize profitability.

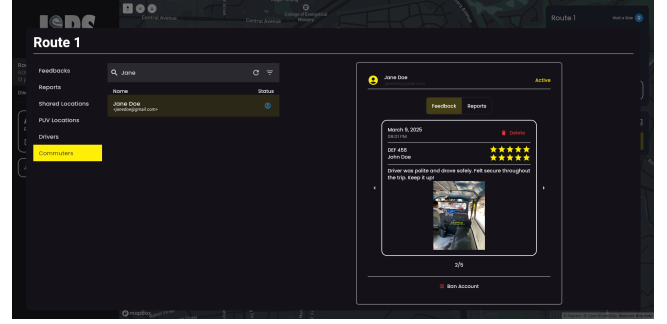


Figure 5: Account Penalty System UI

- (12) **Account Penalty System** - Ensures the platform remains reliable by banning accounts found to be spamming false reports or feedback and can only be re-enabled upon contacting the route managers as shown in Figure 5.

The features of JeePS 2.0 are designed to address key factors that influence passenger satisfaction, including safety, driver's behavior, information material, and service adequacy. The mapping in Table 1 illustrates how each feature aligns with these factors, demonstrating the role of JeePS 2.0 in creating a more efficient and user-friendly transportation system.

Features	Factors
Account System	-
Basic Information	Information Material
Rough Time Estimation	Service Adequacy
Real-Time GPS Location	Service Adequacy
Fare Matrix	Information Material
Passenger Counter	Service Adequacy
Feedback System	Driver's Behavior
Report System	Safety
Passenger Demand System	Information Material
Live Location Sharing	Safety
Passenger Demand Summary	Information Material
Account Penalty System	Information Material

Table 1: Mapping between features of JeePS 2.0 and factors affecting passenger satisfaction

3.2 Hardware Implementation

The JeePS system introduced a manual passenger counter feature to report the available seating capacity of jeepneys as a means of improving the adequacy of the jeepney service and providing information for waiting passengers along the route. However, the results of the user testing show that the feature was inefficient [17]. Thus, in this paper, the authors are proposing a design for implementing a hardware device for a passenger counter, considering the jeepney setup and budget minimization.

APC systems provide a better solution for keeping track of passenger flow, allowing rational use of resources to improve service quality [12]. Some implementations for APCs include a video-based approach utilizing neural networks [15], floor-based sensing [13], and Infrared-based (IR) sensing [14].

However, in a jeepney setting, IR sensing might be the most suitable among the other types of APC. This is in consideration of different factors such as the vehicle's floor-to-ceiling height, seat orientation, step board size, and the balance between cost and accuracy. With side-facing bench seats, video-based APCs might not be practical as they are typically used for vehicles with front-facing seats that provide a clear view of all passengers. Similarly, the low floor-to-ceiling height of the vehicle restricts optimal camera positions. For floor-based sensing, on the other hand, the jeepney's small stepboard size limits the number of sensors that can be installed, negatively impacting accuracy [11]. In contrast, IR-based sensing is still feasible given these factors and is also cheaper than the other types, while maintaining a good accuracy [14][20]. This may also apply to other types of public PUVs, especially those with fewer spatial limitations compared to jeepneys.

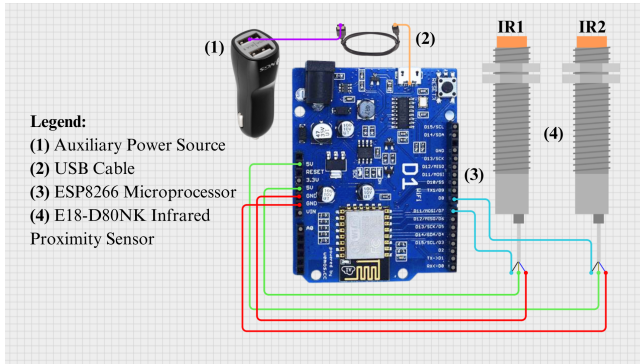


Figure 6: Hardware Schematic Diagram

Figure 6 illustrates the schematic diagram of the hardware system, which consists of only four main components. The ESP8266 Microprocessor (3) is powered by an auxiliary source (1) through a USB micro cable (2), commonly used in vehicles. Both Infrared Proximity Sensors (4) are connected to the 5V and GND pins of the ESP8266 Microprocessor (3), with their signal pins connected to the digital pins D7 and D8.

This design simplifies the circuit, reduces the number of required parts, and minimizes costs while maintaining full functionality. The IR-based APC operates by using two IR sensors to transmit digital signals to the microprocessor. The microprocessor processes these signals and updates the passenger count in the database.

The sequence in which the two IR sensors (IR1 and IR2) are triggered determines whether a passenger is boarding or alighting. IR1 is positioned near the vehicle's exterior, while IR2 is located closer to the interior. Boarding passengers trigger IR1 before IR2, increasing the count, while exiting passengers trigger IR2 before IR1, decreasing the count.

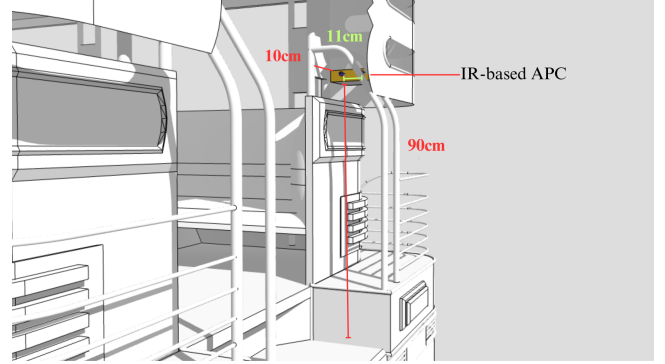


Figure 7: Setup Illustration

As shown in Figure 7, the IR-based APC is mounted 90 cm above the stepboard and 10 cm from the outermost entrance handle with an 11 cm spacing between the IR sensors. This setup reduces incorrect readings compared to placements at the ceiling or near the stepboard, which caused missed detections or unintended counts. A junction box houses the sensors and microprocessor, keeping it safe from accidental bumps. The system is powered by a wired connection to the jeepney's auxiliary power source, reducing the cost for battery maintenance.

4 HARDWARE TESTING RESULTS

Preliminary functional testing for the APC was conducted in a stationary jeepney to ensure the feasibility of its implementation. A total of 120 test runs were conducted to cover different scenarios of passengers entering and exiting the vehicle, each tested 20 times. These included (1) single passengers who entered and exited at a normal pace, (2) multiple passengers who boarded and exited continuously one at a time, (3) simultaneous passengers who entered or exited almost at the same time, (4) hesitant passengers who stepped in but immediately exited, (5) slow-moving passengers who took longer to board or leave, and (6) passengers carrying bags or other accessories that might affect detection. The latency between transmitting the count to the database and when it is displayed in the web application was also measured.

Test Case	False Positives	False Negatives	Correct Detections
Single Passenger	0	0	1.0
Multiple Passenger	0.05	0	0.95
Simultaneous Passengers	0.1	0.2	0.7
Hesitant Passenger	0	0.3	0.7
Slow Movement	0	0.2	0.8
With Obstructions	0	0	1.0

Table 2: IR-based APC Accuracy Testing Result

Table 2 presents the initial hardware testing results. False positives occur when the counter incorrectly updates the passenger count by increasing the count when it should have decreased, or

vice versa. False negatives, on the other hand, happen when a passenger enters or exits the jeepney, but the counter fails to update the count.

The counter demonstrated 100% accuracy for single passengers and passengers carrying items like bags. The accuracy remained high at 95% for multiple passengers boarding one after another. However, accuracy dropped to 80% for slow-moving passengers since lingering in the sensor's range sometimes caused missed detections. Both hesitant and simultaneous passengers had the lowest accuracy at 70%, as hesitation at the entrance or people entering close together made it harder for the sensors to detect them correctly. This may be due to the APC's limitation in concurrent detection, where both sensors are triggered almost simultaneously.

Across all test cases, the system achieves an average accuracy of 85.83%, with potential for further improvement.

Moreover, the latency between the database's count update and its display within the system was evaluated using the single-passenger scenario. The results show a minimum delay of 0.03s, a maximum delay of 1.07s, and an average delay of 0.033s, indicating minimal latency. This suggests that the system operates in near real-time and ensures timely updates necessary for tracking passenger count.

However, actual performance may vary in real-world conditions. Factors such as network inconsistencies in certain areas may lead to greater latency between passenger count updates and their display. Additionally, passengers carrying large objects, such as boxes or luggage, could result in miscounts. Moreover, harsh weather conditions, particularly heavy rain, may pose a risk of water penetration, potentially damaging the APC.

5 PROPOSED TESTING PLAN

5.1 Functional Testing

Functional testing will be conducted to verify that the features work according to the specified requirements. This will serve as the alpha testing of the web and mobile applications. Each evaluator will assume the roles of different users, namely guests, passengers, drivers, and route managers. They will be given a list of features, including both the original and improved features of JeePS, along with their expected behavior. This will ensure that no changes were made to the original features and that the improved features work as intended in different use cases.

5.2 Field Testing

Field testing will be conducted to assess the effectiveness of the web and mobile applications in a real-world environment. This will take place for two weeks, where JeePS will be tested during the first week, followed by JeePS 2.0 in the second week.

Testing will involve drivers who will be tasked with completing the SUS questionnaire [2] and will be interviewed to gather further insights on their experience with the app. In addition, at least 30 selected commuters will answer the same SUS questionnaire [2] and an effectiveness questionnaire at the end of each week. The same group of commuters will answer the questionnaires for both weeks. The effectiveness questionnaire is a Likert scale-based survey based on the study by Ong et al. [16], where each question targets a factor affecting passenger satisfaction. The questionnaire must focus on

the improvements made to JeePS while ensuring that the questions remain applicable to both versions to avoid bias.

The results of the effectiveness questionnaire will be evaluated using a paired t-test to determine if there is a significant difference in passenger satisfaction between JeePS and JeePS 2.0.

6 DOMAIN EXPERT VALIDATION

To validate the features of JeePS 2.0, an interview with Dr. John Justine Villar, an expert in transportation and technology from the National Center for Transportation Studies (NCTS), was conducted. He particularly appreciated that the APC is not disruptive, which addresses the major issue in the previous version of JeePS where the manual implementation posed risks to drivers' safety. He also agreed that the IR-based sensing is the most suitable implementation for APC as video-based and floor-based sensing are expensive and require significant resources, which could lead to drivers being hesitant in installing them in jeepneys. Additionally, he noted that the APC is acceptable as long as it covers the typical scenarios such as single and multiple passengers entering or exiting the jeepney. While it may not be entirely accurate for rare cases such as when passengers hesitate while entering the jeepney, it can be further improved with more calibration and testing.

Regarding the passenger demand summary, the domain expert highlighted the usefulness of historical data for drivers in planning their trips. He also acknowledged the importance of providing extra evidence such as images to verify the validity of feedback and reports.

Overall, Dr. John Justine Villar validated that JeePS 2.0 covers the necessary features and addresses the challenges faced by the original JeePS.

7 CONCLUSION AND RECOMMENDATIONS

The enhancements introduced in JeePS 2.0 aim to address the limitations of the original system by improving both its software and hardware components. To resolve the limitations identified in JeePS, several software enhancements were implemented in both the web and mobile applications. The system now supports persistent real-time GPS updates, an extended fare matrix, and moderated feedback and report system. Additionally, the introduction of live location sharing and account penalty system are designed to ensure reliability and improve passenger safety. Furthermore, the passenger demand summary is intended to support drivers and route managers in decision making and better route scheduling.

On the hardware side, the integration of an IR-based APC resolves the inefficiencies and safety risks of manual counting. Preliminary testing results show an average accuracy of 85.83% which demonstrates the system's reliability in tracking passenger occupancy. Even with this accuracy level, the collected data can still be valuable for other applications like a passenger demand forecast, where the focus is on the overall passenger count trend rather than precision on individual passenger detection. Furthermore, the domain expert from NCTS validated the practicality and feasibility of this solution for jeepney drivers.

These improvements are expected to contribute to passenger satisfaction and service quality of jeepneys, but their actual impact will be determined once the proposed testing plan is conducted.

The testing plan validated by the domain expert includes functional testing to ensure that all features meet the intended requirements and field testing to evaluate real-world usability through SUS assessment and an effectiveness questionnaire. A paired t-test will determine whether JeePS 2.0 significantly improves passenger satisfaction compared to the previous version. The results of these evaluations will provide insights into how well JeePS 2.0 enhances passenger experience and service quality.

JeePS 2.0 covers the necessary features to potentially improve passenger satisfaction and service quality of jeepneys. While the APC effectively handles the typical scenarios of passengers entering and exiting jeepneys, it is still recommended to address rare cases such as multiple passengers boarding simultaneously, hesitant passengers, and slow-moving passengers. Further calibrating the sensitivity of the IR sensors and adjusting the delay between detecting passengers, along with considering alternative placements for different jeepney setups can be done to improve overall accuracy. Future developments could also expand the system to other types of PUVs such as modern jeepneys and buses, increasing its applicability across different modes of public transportation.

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